

MUHAMMAD ARABI

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OBJECTIVE

Software Engineering student at COMSATS University Islamabad seeking a summer internship in software or IT development. Experienced in app development, web development, database management, and system design through six academic projects

EDUCATION

Comsats University Islamabad

Bachelor of Science in Software Engineering

2024 – 2028

Islamabad, Pakistan

CGPA: 3.19/4.0

PROJECTS

NovaGrid - Solar Solutions Platform | *HTML, CSS, JavaScript, PHP, MySQL*

- Built a full-stack multi-role web platform with separate portals for customers, admins, and public users across 15+ interconnected pages.
- Implemented a live Solar Load Estimator with real-time kWh calculations, system sizing recommendations, and ROI projections based on user appliance input.
- Designed a RESTful PHP API layer with endpoints for authentication, lead management, inventory, and project status updates.
- Developed an admin dashboard featuring Chart.js data visualizations including revenue trends, lead funnels, inventory levels, and project Kanban boards.
- Applied a dual light/dark theming system using CSS custom properties and ensured full mobile responsiveness across all pages.

ATM Software with GUI | *Java, Swing, OOP*

- Developed a desktop banking application in Java with a fully functional Swing GUI supporting deposits, withdrawals, and balance inquiry.
- Applied core OOP principles including Inheritance, Polymorphism, and Encapsulation to ensure modular and maintainable code architecture.

Facial Emotion Recognition System | *Python, DCNN, TensorFlow*

- Developed a Deep Convolutional Neural Network (DCNN) to accurately detect and classify human facial emotions across multiple expression categories.
- Implemented probability theory and swarm optimization techniques to improve model accuracy over baseline benchmarks.
- Preprocessed and augmented large image datasets to reduce overfitting and improve neural network generalization.

Weather App | *Flutter, Dart, REST API*

- Integrated OpenWeatherMap REST API to fetch and display real-time weather data including temperature, humidity, and forecasts for searched cities.
- Designed a responsive UI using custom Flutter widgets including Cards, ListTiles, and animated loading states for smooth user experience.
- Handled asynchronous API calls using `async/await` and implemented error handling for network failures and invalid city inputs.

Expense Tracker | *Flutter, Dart, Provider*

- Implemented Provider pattern for centralized state management, ensuring real-time UI updates across screens without unnecessary widget rebuilds.
- Built dynamic transaction lists using `ListView.builder` for memory-efficient rendering of variable-length expense data.
- Persisted user data locally using asynchronous storage, maintaining full expense history across app sessions.

Railway Reservation System | *C++, DSA, Linked Lists*

- Designed a dynamic booking system using Linked Lists to manage passenger records and reservations efficiently.
- Implemented Queue and Stack data structures to handle ticket waiting lists and enforce correct booking processing order.
- Optimized search algorithms to enable real-time seat availability checks across multiple train routes.

TECHNICAL SKILLS

Languages: Python, Java, C++, Dart, JavaScript, HTML, CSS, SQL

Frameworks & Libraries: Flutter, TensorFlow, NumPy, Pandas, Matplotlib

Web Development: PHP, MySQL, REST APIs, Chart.js, Responsive Design

AI & Data Science: Machine Learning, Deep Learning, Computer Vision

Developer Tools: Git, GitHub, VS Code, IntelliJ IDEA

Core Concepts: Data Structures & Algorithms, OOP, Database Management Systems, Operating Systems, Software Engineering, Software Design & Architecture, Software Requirement Engineering

VOLUNTEER EXPERIENCE

Google Developer Group on Campus (GDGoC) CUI

Islamabad, Pakistan

Mobile Development Volunteer

2024 – Present

- Volunteering in the mobile development domain, attending Flutter workshops and collaborating with peers on app development initiatives.

IEEE Student Branch - COMSATS Islamabad

Islamabad, Pakistan

Volunteer

2024 – Present

- Assisted in organizing 3+ technical seminars and inter-university collaborative tech events, coordinating logistics and speaker sessions.

CERTIFICATIONS

Google IT Automation: Crash Course on Python - Google (Coursera)

Programming for Everybody (Getting Started with Python) - University of Michigan (Coursera)

Introduction to SQL - DataCamp

Understanding Data Science - DataCamp